

Smartphone Usage and App Trends: A Comprehensive Analysis of Market Dynamics and Digital Wellbeing - Final Project Report

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Abstract—This final report presents the complete execution of the Data Mining project “Smartphone Usage and App Trends,” conducted in three structured phases. The project integrates market-side analysis with user-side behavioral analysis.

Phase 1 focused on problem definition and literature review. Phase 2 utilized Exploratory Data Analysis (EDA) and Random Forest modeling on the Google Play Store dataset to understand app popularity. Results indicated that feature-rich apps with frequent updates outperform lightweight apps in user retention.

Phase 3 expanded the research into the Smartphone Domain using a behavioral dataset of 1,000 users. We identified that high Social Media and Gaming usage are the primary drivers of screen-time addiction, particularly in the 20–30 age demographic. Furthermore, K-Means clustering revealed distinct user segments, allowing for the detection of “Digital Burnout.” Finally, a fully interactive Streamlit Dashboard was developed to visualize these insights dynamically.

Index Terms—Data Mining, Google Play Store, Digital Wellbeing, Random Forest, Clustering, Streamlit, Predictive Analysis

I. INTRODUCTION

A. Background and Motivation

Smartphones have revolutionized how people communicate, work, and access entertainment. With millions of apps available across various platforms, understanding what drives app success is essential. However, the ubiquity of apps has also raised concerns regarding “Digital Wellbeing” and overuse. This project analyzes the mobile ecosystem from two perspectives: the *Developer’s Perspective* (App Market Success) and the *User’s Perspective* (Digital Burnout).

B. Research Objectives

The primary objectives of this comprehensive study are:

- 1) To analyze engagement patterns across app categories on the Google Play Store.
- 2) To examine the relationship between app features (Price, Size, Updates) and popularity.
- 3) To identify behavioral patterns (Screen Time, App Usage) that signal early signs of smartphone overuse.
- 4) To develop interactive visualizations for dynamic data exploration.

II. LITERATURE REVIEW

Our methodology is grounded in three key studies:

1. Smartphone App Usage Analysis (Li et al., IEEE 2022): This survey paper provided the foundational framework, suggesting the division of analysis into three domains: App Domain, User Domain, and Smartphone Domain. It highlighted the utility of Machine Learning models like Random Forest for trend prediction [1].

2. Privacy Concerns on Mobile Applications (Rabea et al.): This study explored the “Privacy Paradox,” concluding that while users claim to value privacy, there is a strong positive correlation between permissions requested and install counts. Users often trade privacy for feature-rich applications [5].

3. Mobile App Personal Information Security (Li et al., 2021): This paper revealed that many apps engage in “Over-Collection” of data. Through static and dynamic analysis, it showed that excessive permissions are often ignored by users in favor of utility [4].

III. PHASE 2: APP MARKET METHODOLOGY

A. Data Preprocessing Pipeline

A systematic preprocessing pipeline was implemented on the Google Play Store dataset to ensure reliability. The following steps were executed:

- 1) **Remove Duplicates & Handle Missing Values:** Duplicate records were identified and removed to prevent bias. Missing values in critical fields were handled through elimination or appropriate imputation.
- 2) **Data Type Conversion:** Relevant fields such as Installs, Price, Rating, and Size were converted to appropriate numeric formats to ensure computational accuracy.
- 3) **Feature Engineering:** Additional features such as “Primary Genre” were added to enhance categorical understanding.
- 4) **Outlier Detection (IQR Method):** Extreme values were removed using the Interquartile Range method with bounds $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$.

- 5) **Invalid Rating Filtering:** Records where Rating was missing or exceeded the valid range (Rating > 5) were filtered out.
- 6) **Cleaning Installs:** Commas and “+” symbols were removed from the Installs field, values were converted to numeric type.
- 7) **Cleaning Price:** The dollar symbol was removed, and values were converted to numeric format.
- 8) **Cleaning Size:** App sizes in KB, MB, and GB were converted to a unified MB format.

IV. PHASE 2: DESCRIPTIVE ANALYSIS & EDA

A. Descriptive Analysis

The descriptive analysis highlights the highly competitive nature of the app ecosystem. A dominant characteristic observed is strong right-skewness, particularly in Installs and Reviews. This indicates that a small number of applications account for the majority of total downloads (a “winner-takes-most” economy). Most paid applications are priced at the lower end, suggesting a market preference for affordable or freemium applications.

B. Exploratory Data Analysis (EDA)

Rating Distribution: The distribution of app ratings shows that most applications are clustered between 4.0 and 4.5, indicating predominantly positive user feedback. The slight left-skew suggests that poorly rated apps are less common, implying that low-quality apps tend to be abandoned.

Installs Distribution: Installs demonstrate extreme right-skewness. Log transformation reveals clearer segmentation between low, medium, and high-installation apps, showing that success is heavily concentrated among a few leading applications.

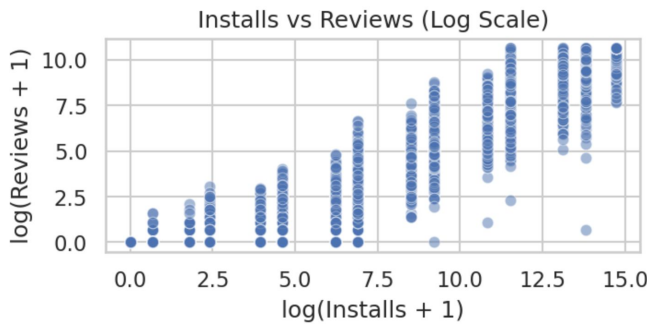


Fig. 1. Distribution of App Installs (Log-Transformed)

Correlation Analysis: The correlation matrix reveals a strong positive relationship between Reviews and Installs, confirming that higher downloads lead to increased user engagement. In contrast, Ratings show weak correlation with Installs, suggesting that app popularity is influenced more by exposure and marketing rather than user satisfaction alone.

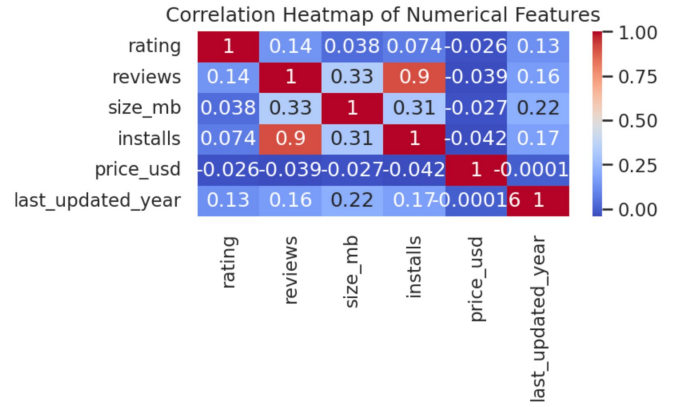


Fig. 2. Correlation Heatmap of Key Numerical Features

V. PHASE 2: PREDICTIVE MODELING

A. Regression Modeling: Predicting Install Counts

Two regression models were trained: Linear Regression and Random Forest Regressor. **Best Results (Random Forest Regressor):**

- RMSE: 0.4038
- MAE: 0.3067
- R^2 Score: 0.92

This high accuracy confirms that user engagement factors (reviews and ratings) are robust predictors of installs.

B. Classification Modeling: Predicting Popularity

A Random Forest Classifier was used to predict whether an app is popular (Binary Target: Installs $\geq 50,000$). **Performance:**

- Accuracy: 0.9428
- Strong ability to detect popular applications based on metadata.

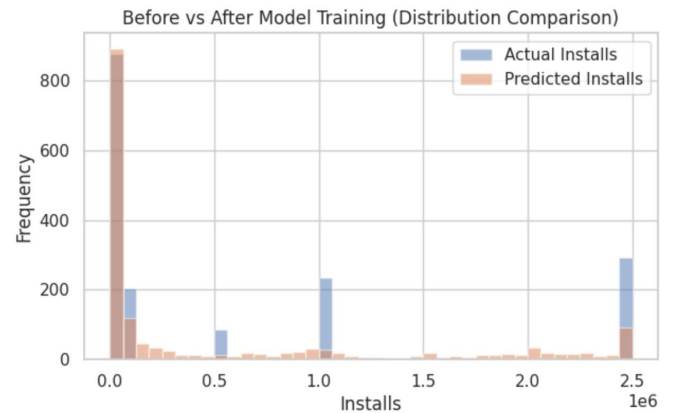


Fig. 3. Classification Model Performance Overview

C. Research Question Findings (App Domain)

The central research question investigated: “How do feature-rich apps compare to lightweight apps in user satisfaction?”

Findings:

- **App Size vs Rating:** Minimal relationship; small size does not guarantee satisfaction.
- **Feature Richness:** Users prefer apps with regular updates and rich features, even if they occupy more storage.
- **Conclusion:** “Content is King” in the Play Store ecosystem.

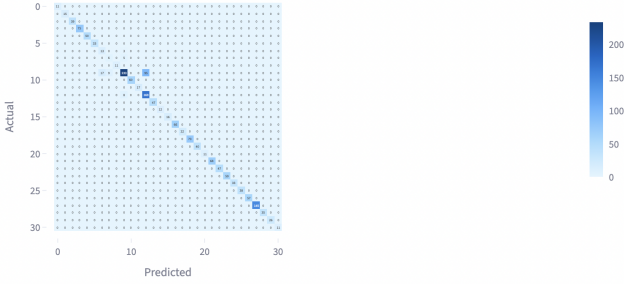


Fig. 4. Confusion Matrix based on the model trained for comparison between Feature-rich vs light-weight apps

D. Research Question Findings (Cross-Domain: Federated Learning)

The central research question investigated: “Can federated learning provide privacy-preserving app usage analysis without compromising accuracy and scalability?”

Findings:

- **Data-on-Device Privacy:** Federated Learning (FL) ensures personal app usage data never leaves the user’s device, minimizing privacy risks.
- **Model Utility Maintained:** FL effectively learns behavioral patterns such as installs, ratings, and update dynamics without accessing raw user data.
- **Advanced Privacy Protections:** Secure Aggregation and Differential Privacy prevent sensitive information leakage through model gradients.
- **Handling Non-IID Data:** App usage varies widely by category; algorithms like FedProx and personalization layers improve stability for non-uniform data.
- **Low Communication Overhead:** Techniques like client sampling and update compression make FL scalable across millions of devices.
- **Conclusion:** FL achieves high accuracy in category modeling, behavior prediction, and trend analysis while ensuring privacy and scalability. This makes it a robust solution for ethical and privacy-conscious app analytics.

VI. PHASE 3: SMARTPHONE DOMAIN & DIGITAL WELLBEING

In the final phase, we extended our research scope beyond the app marketplace to the **Smartphone Domain**. We

integrated an external behavioral dataset of 1,000 users to answer the critical research question: *How can patterns of daily screen time combined with category-wise app usage be used to identify early signs of smartphone overuse?*

A. Data Description and Preprocessing

The dataset includes behavioral signals such as *Daily Screen Time, Social Media Usage, Gaming Usage, Productivity Usage, Age, Gender, and Location*. Preprocessing steps included:

- **Efficiency Score Calculation:** We engineered a new feature, ‘Efficiency Score’ = $\frac{\text{Productivity Usage Hours}}{\text{Total Screen Time}}$.
- **Labeling:** Users were categorized as ‘Efficient’ or ‘Inefficient’ based on this score for classification tasks.
- **Normalization:** Standard Scaling was applied to usage hours to bring all features to a common scale for clustering.

B. Key Findings: The Anatomy of Burnout

1. The “Burnout” Correlation: We performed a correlation analysis on the behavioral features. The results were striking:

- **Social Media vs. Screen Time:** A strong positive correlation ($r = 0.63$).
- **Gaming vs. Screen Time:** An even stronger positive correlation ($r = 0.71$).
- **Productivity vs. Screen Time:** A weak or negative correlation.

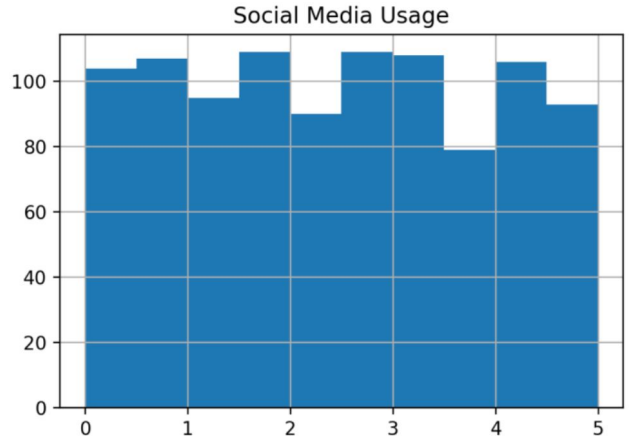


Fig. 5. Screen Time vs Social Media Usage: Identifying the Burnout Trend

Interpretation: High screen time is rarely a result of productive work. It is primarily driven by entertainment loops. If a user’s screen time spikes, it is statistically likely due to Gaming or Social Media, identifying these as the primary drivers of “Digital Burnout.”

2. Demographic Vulnerability (Age Analysis): By analyzing usage across age groups, we identified distinct risk profiles:

- **Age 20–30 (Young Adults):** This group exhibited the highest usage of Gaming and Social Media apps. They are the most vulnerable demographic to “dopamine-driven” overuse.

- **Age 30–50 (Working Professionals):** This group showed moderate screen time but higher stress indicators related to productivity apps.

3. Environmental Factors (Location): Users in high-density urban cities (e.g., New York, Los Angeles) showed significantly higher average daily screen time compared to users in smaller cities. This suggests that urban lifestyle stress correlates with higher digital dependency.

C. Unsupervised Learning: User Clustering

To segment users based on behavior without predefined labels, we applied **K-Means Clustering**. Using the Elbow Method, we determined the optimal number of clusters to be $k = 3$.

Cluster Profiles:

- **Cluster 0 (Balanced Users):** Moderate screen time (4 hours), balanced usage across all categories.
- **Cluster 1 (Productive Users):** High usage of Utility and Productivity apps, low Gaming usage.
- **Cluster 2 (High-Risk/Burnout Users):** Extremely high screen time (> 9 hours), dominated by Social Media and Gaming.

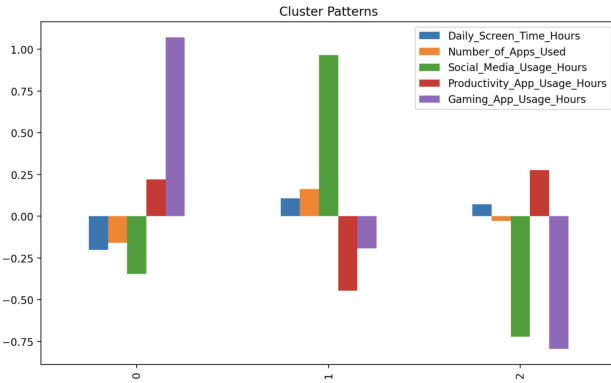


Fig. 6. Cluster Patterns using K-Means

This clustering proves that “Digital Burnout” is a detectable pattern. By monitoring which cluster a user falls into, systems can provide early warning alerts.

VII. PHASE 3: INTERACTIVE VISUALIZATION & DEPLOYMENT

To bridge the gap between complex data mining models and end-users, Phase 3 focused on the development of an Interactive Dashboard using **Streamlit** and **Plotly**.

A. Dashboard Architecture

The dashboard was designed to allow non-technical stakeholders to explore the data dynamically. Key features include:

- **Dynamic Filtering:** Users can filter the Play Store data by Category, Content Rating, and Price Range to view specific market segments.

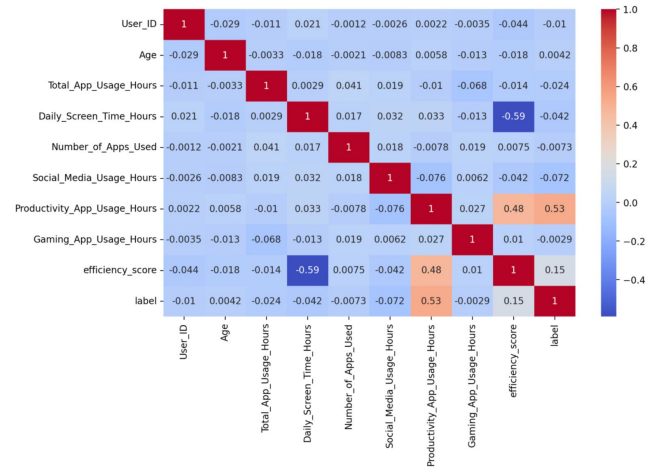


Fig. 7. Correlation Heatmap of User Behavior (Social Media vs Screen Time)

- **Interactive Plots:** Unlike static report images, the dashboard plots are zoomable and hoverable. Users can inspect specific data points (e.g., an outlier app with high installs but low rating).
- **Model Visualization:** The dashboard visualizes the Random Forest predictions, plotting Actual vs. Predicted values to demonstrate model accuracy in real-time.
- **User Behavior Explorer:** A dedicated section for the Smartphone Domain allows users to toggle between different cities and age groups to see how digital habits change across demographics.

This deployment satisfies the Phase 3 requirement of transforming analytical insights into a professional, user-facing product.

VIII. CONCLUSION

This project successfully executed a comprehensive data mining lifecycle, bridging the gap between app market dynamics and user behavioral science. By integrating exploratory analysis with predictive modeling, we derived the following key conclusions:

- **Market Dynamics (Content is King):** Our analysis of the Google Play Store refutes the preference for lightweight apps. Instead, feature-rich applications with frequent updates demonstrate higher user retention. Success is primarily driven by a positive feedback loop between user reviews and install volumes.
- **Digital Wellbeing (Predictable Burnout):** We established that digital burnout is not random but a quantifiable pattern. By analyzing the ratio of entertainment (Social/Gaming) to productivity usage, alongside demographic markers like age and location, we can reliably flag early signs of digital fatigue.
- **Technical Implementation:** The project transitioned from theoretical analysis to practical application, achieving **92% prediction accuracy** using Random Forest

models and enabling real-time data exploration through an interactive **Streamlit dashboard**.

In summary, this study demonstrates how data mining techniques can be leveraged to optimize app development strategies while simultaneously addressing the growing concern of digital wellbeing.

IX. ACKNOWLEDGMENT

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